

Yinxi Li (Larry)

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Education

University of Waterloo, Canada

May 2025 – April 2027

M.Math. in Computer Science, Advisor: [Pengyu Nie](#)

- **Research Interests:** LLMs, AI4SE
- **Coursework:** Machine Learning for SE, Computational Linguistics, Mining & Analysis of SE Data
- GPA: 97.7/100

The Chinese University of Hong Kong, Hong Kong (SAR)

Sept 2020 – July 2024

B.Sc. in Computer Science

- **ELITE Stream** [🔗](#): A selective honors program requiring exceptional performance in higher mathematics and computer science, including graduate-level coursework and rigorous technical prerequisites.
- **Coursework:** Foundations of Optimization, Machine Learning Theory, Fundamentals of AI

ETH Zürich, Switzerland

Feb 2023 – Sept 2023

Exchange Student

- **Coursework:** Rigorous Software Engineering, Introduction to ML, Data Modelling and Databases

Project Experience

Unit Test Transplantation with Retrieval-Augmented Generation

Sept 2024 – Dec 2024

- Developed a RAG pipeline for test transplantation to improve unit test generation in API frameworks
- Analyzed model performance on 2,000+ unit test cases extracted from 7 diverse open-source repositories, identifying optimization opportunities and proposing enhancement strategies

Optimizing Structure of Vision-Language Retrieval System

Aug 2023 – June 2024

ELITE Stream Final Year Project, Supervisor: [Eric Lo](#)

- Developed and integrated a proxy model and optimized retrieval structures, achieving up to 50% faster object detection and 1000× inference speedup with minimal accuracy loss
- Designed a vision-language retrieval system for large-scale video datasets, enabling efficient and diverse top-K results with reduced computational complexity from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd)$
- Conducted comprehensive evaluations, including baseline comparisons and ablation studies, demonstrating superior retrieval accuracy, diversity, and architectural effectiveness

Adaptive Streaming for Short-Video Services

June 2022 – Aug 2022

CUHK Summer Research, Supervisor: [Lee Yiu-Bun](#)

- Implemented and evaluated multiple adaptive bitrate (ABR) algorithms (including BBA, BOLA, and the RL-based Pensieve), and tuned the simulator to support focused testing on short video samples

Competition

World University Students' Digital Intelligence Competition Invitational Tournament (by Tencent)

April 2023 – July 2023

Ranked first among overseas institutions, Mentor: [Tan LEE](#)

- Developed a multi-agent reinforcement learning framework for Honor of Kings, with optimized reward functions and customized agent strategies

Research Internship Experience

Research Assistant

Remote

Southern University of Science and Technology, Supervisor: [SONG Xuan](#)

Aug 2024 – Oct 2024

- Developed an expert knowledge collaboration system for the automotive quality inspection value chain based on large language model and retrieval-augmented generation

Research Intern

Research Institute of Tsinghua University in Shenzhen

*Shenzhen, China
April 2024 - June 2024*

- Implemented statistical, ML, and deep learning methods for multivariate time-series anomaly detection
- Utilized a variational autoencoder for anomaly detection, improving customer load monitoring

Research Intern

Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences

*Changchun, China
June 2020 - Aug 2020*

- Optimized object detection of moving objects in specific directions in videos

Research Community Service

Teaching Assistant: CS 246, CS346, CS446 (25 Spring, Fall, 26 Winter, UWaterloo)

Sub-Reviews: ASE 2025, ARR Oct 2025

Association Activities Experience

CUHK Mainland Undergraduate Association (19th & 20th cabinet)

Aug 2021 - July 2023

- Principal of Information Department: Managed communication channels
- Assistant of Sports Department: Coordinated Sports Festival and e-sports tournaments

CUHK Mainland Undergraduate Association Orientation Camp

Aug 2021 - July 2023

- Group Leader(2021-23): Mentored 7 groups and supported 60+ freshmen in academic and campus life

Additional Information

Open Source Contributions: [TokDrift](#) , [Arxiv Paper Agent](#) 

Programming Languages: Python, Java, Shell Script